

Course 6033B

Semester VI

Theory

4 Credits

Embedded System Design with PIC

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electrical & Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6033B as Embedded System Design with PIC in Semester VI for Electrical & Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Introduction to Embedded System Design course and polytechnic PIC microcontroller curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Embedded designs, PIC programs, circuit schematics and system safety must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support PIC18F architecture, C programming, MPLAB X IDE, and peripheral interfacing. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Embedded System Design with PIC studies the architecture, programming and application of PIC microcontrollers in embedded systems. It develops the ability to analyze PIC18F architecture, write embedded C programs using MPLAB X IDE, interface sensors and actuators, and design compact control solutions while respecting the technical and safety boundaries of electronic and embedded engineering.

Malayalam support: □ handbook □□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□□.

Prerequisite foundation

Digital electronics, C programming, basic microprocessor architecture and analog electronics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the architecture, memory organization and internal peripherals of PIC microcontrollers.
2. Apply embedded C programming to implement control logic and peripheral drivers for PIC devices.
3. Analyze the interfacing requirements for sensors, displays and motors in PIC-based systems.
4. Explain the systematic design process for embedded systems including hardware-software co-design.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO ഉള്ളത് exam-ൽ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ. Define, explain, apply ഉള്ളത് action words ഉപയോഗിക്കാൻ.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the architecture and internal peripherals of the PIC18F microcontroller.	Understand
CO2	Develop embedded C programs for PIC microcontrollers using MPLAB X IDE.	Apply
CO3	Interface PIC microcontrollers with sensors, LCDs, keypads and motor drivers.	Apply
CO4	Design and simulate simple embedded systems for industrial and consumer applications.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	PIC Microcontroller Architecture	Introduction to PIC microcontrollers; PIC18F architecture: CPU, Harvard architecture, pipelining; Memory organization: Flash (Program), SRAM (Data), EEPROM; Special Function Registers (SFRs); Oscillator and Reset.	Approx. 14 hours	CO1	Module 1
2	Embedded C Programming for PIC	Embedded C fundamentals; MPLAB X IDE and XC8 compiler; Configuration bits; I/O programming; Timers and Counters; Interrupt handling in PIC18F.	Approx. 16 hours	CO2	Module 2
3	Peripheral Interfacing and Control	Interfacing: LED, 16x2 LCD, 4x4 Keypad; Analog-to-Digital Converter (ADC) module; CCP module: PWM for DC motor control; Sensors: Temperature (LM35), Ultrasonic.	Approx. 16 hours	CO3	Module 3
4	Embedded System Design and Simulation	Serial communication: UART (RS232); Basics of I2C and SPI; Hardware-software co-design; Proteus simulation; Debugging techniques (ICD); Simple project: Digital thermometer or Speed controller.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

PIC Microcontroller Architecture

Introduction to PIC microcontrollers; PIC18F architecture: CPU, Harvard architecture, pipelining; Memory organization: Flash (Program), SRAM (Data), EEPROM; Special Function Registers (SFRs); Oscillator and Reset.

CO1 · Approx. 14 hours

Embedded C Programming for PIC

Embedded C fundamentals; MPLAB X IDE and XC8 compiler; Configuration bits; I/O programming; Timers and Counters; Interrupt handling in PIC18F.

CO2 · Approx. 16 hours

Peripheral Interfacing and Control

Interfacing: LED, 16x2 LCD, 4x4 Keypad; Analog-to-Digital Converter (ADC) module; CCP module: PWM for DC motor control; Sensors: Temperature (LM35), Ultrasonic.

CO3 · Approx. 16 hours

Embedded System Design and Simulation

Serial communication: UART (RS232); Basics of I2C and SPI; Hardware-software co-design; Proteus simulation; Debugging techniques (ICD); Simple project: Digital thermometer or Speed controller.

CO4 · Approx. 14 hours

MODULE 1

PIC Microcontroller Architecture

Introduction to PIC microcontrollers; PIC18F architecture: CPU, Harvard architecture, pipelining; Memory organization: Flash (Program), SRAM (Data), EEPROM; Special Function Registers (SFRs); Oscillator and Reset.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

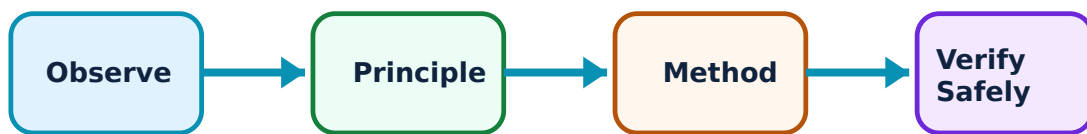
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PIC Microcontroller Architecture: identify the system, state its principle, follow the correct method and verify the result safely.

1. PIC internal structure–

PIC (Peripheral Interface Controller) microcontrollers use a Harvard architecture with separate buses for instructions and data. The PIC18F series features an 8-bit CPU, multi-level hardware stack, and integrated peripherals. Pipelining allows fetching and executing instructions simultaneously.

Study and application method

Sketch the block diagram of the PIC18F microcontroller; explain the advantage of 'Harvard Architecture' over Von Neumann.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of the PIC18F microcontroller; explain the advantage of 'Harvard Architecture' over Von Neumann.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ ചോദ്യം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. ഈ diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഈ result ഈ application ഈ ഉപയോഗിക്കണം. Technical terms English-ൽ ഉപയോഗിക്കണം.

Common mistake / precaution: PIC microcontrollers are sensitive to ESD. Do not handle the ICs without authorised grounding and anti-static precautions.

2. Memory and SFRs–

Program memory stores code in Flash, while Data memory uses SRAM and EEPROM. Special Function Registers (SFRs) like TRIS, PORT, and LAT are used to control I/O and internal modules. The 'Access Bank' simplifies register addressing.

Study and application method

Describe the role of 'TRIS' and 'PORT' registers in I/O control; explain the purpose of the 'EEPROM' in embedded systems.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions,

and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of 'TRIS' and 'PORT' registers in I/O control; explain the purpose of the 'EEPROM' in embedded systems.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നോക്കുക. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കുക. ഏതു result ലഭിക്കും. ഏതു application ഉപയോഗിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Common mistake / precaution: Improper memory mapping or stack overflow can cause system crashes. All memory-intensive operations must follow authorised software design guidelines.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Embedded C Programming for PIC

Embedded C fundamentals; MPLAB X IDE and XC8 compiler; Configuration bits; I/O programming; Timers and Counters; Interrupt handling in PIC18F.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

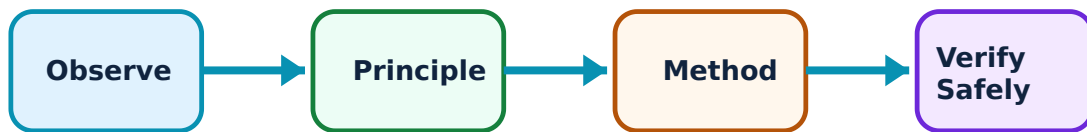
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Embedded C Programming for PIC: identify the system, state its principle, follow the correct method and verify the result safely.

1. Software development environment–

MPLAB X IDE is the official development platform for PIC. The XC8 compiler converts C code into machine-level hex files. 'Configuration Bits' are used to set the oscillator type, watchdog timer, and power-up timer before code execution.

Study and application method

Explain the steps to create a new project in 'MPLAB X IDE'; write a C code snippet to configure the internal oscillator.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the steps to create a new project in 'MPLAB X IDE'; write a C code snippet to configure the internal oscillator.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure നൽകിയിരിക്കുന്നു. ഏതു result നൽകിയിരിക്കുന്നു application നൽകിയിരിക്കുന്നു. Technical terms English-ൽ നൽകിയിരിക്കുന്നു.

Common mistake / precaution: Incorrect configuration bits can prevent the microcontroller from starting. Do not program configuration bits without authorised verification of the target hardware setup.

2. Timers and Interrupts–

Timers are used for generating precise delays and measuring time intervals. Interrupts allow the CPU to respond to external or internal events immediately. PIC18F supports high and low priority interrupts for efficient task management.

Study and application method

Write a C program to generate a 1-second delay using 'Timer0'; explain the difference between 'Polled I/O' and 'Interrupt-driven I/O'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write a C program to generate a 1-second delay using 'Timer0'; explain the difference between 'Polled I/O' and 'Interrupt-driven I/O'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Uncontrolled interrupt nesting can lead to stack overflow. All interrupt service routines (ISRs) must be kept short and follow authorised real-time design rules.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Peripheral Interfacing and Control

Interfacing: LED, 16x2 LCD, 4x4 Keypad; Analog-to-Digital Converter (ADC) module; CCP module: PWM for DC motor control; Sensors: Temperature (LM35), Ultrasonic.

Approx. 16 hours

CO3

Understand · Apply

Learning goals

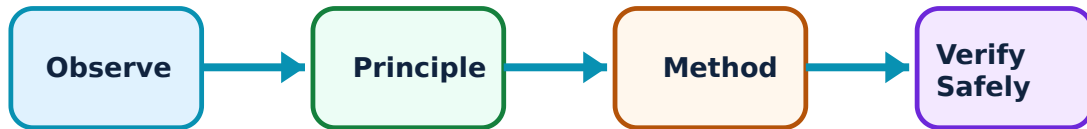
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Peripheral Interfacing and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. User interface and sensors–

Interfacing LCDs and keypads allows for user interaction. The ADC module converts analog sensor voltages (e.g., from an LM35 temperature sensor) into digital values for processing. Proper voltage scaling and reference selection are essential for accuracy.

Study and application method

Sketch the circuit for interfacing a '16x2 LCD' with PIC18F in 4-bit mode; write a C function to read a value from the 'ADC'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the circuit for interfacing a '16x2 LCD' with PIC18F in 4-bit mode; write a C function to read a value from the 'ADC'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure നൽകിയിരിക്കുന്നു. ഏതു result നൽകിയിരിക്കുന്നു application നൽകിയിരിക്കുന്നു. Technical terms English-ൽ എഴുതിക്കുക.

Common mistake / precaution: Sensor signals are prone to noise. Do not interface sensors without authorised signal conditioning, filtering, and shielded cables.

2. Actuator control using PWM–

The Capture/Compare/PWM (CCP) module is used to generate PWM signals. PWM is essential for controlling the speed of DC motors and the brightness of LEDs. H-bridge drivers like L293D are used to interface motors with the microcontroller.

Study and application method

Explain the principle of 'Pulse Width Modulation' (PWM); sketch the interface between a PIC18F and a 'DC Motor' using a driver IC.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of 'Pulse Width Modulation' (PWM); sketch the interface between a PIC18F and a 'DC Motor' using a driver IC.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. diagram / circuit / formula / procedure വ്യക്തമായി വിശദീകരിക്കുക. ഫോർമുല, പ്രോസീഡർ, റിസൾട്ട്, അപ്ലിക്കേഷൻ എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. Technical terms English-ൽ ഉൾപ്പെടെയുള്ളവ.

Common mistake / precaution: Motors generate inductive back-EMF. Do not connect motors to the microcontroller without authorised flyback diodes and isolation circuits.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Embedded System Design and Simulation

Serial communication: UART (RS232); Basics of I2C and SPI; Hardware-software co-design; Proteus simulation; Debugging techniques (ICD); Simple project: Digital thermometer or Speed controller.

Approx. 14 hours

CO4

Understand · Apply

Learning goals

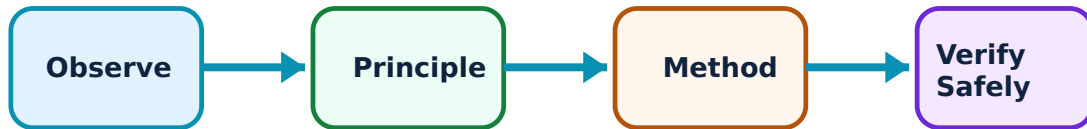
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Embedded System Design and Simulation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Communication protocols–

UART is used for simple point-to-point serial communication (e.g., with a PC). I2C and SPI are used for communication between the microcontroller and multiple peripheral ICs like RTCs, sensors, and external memory.

Study and application method

Explain the role of 'TX' and 'RX' pins in UART communication; compare 'I2C' and 'SPI' in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the role of 'TX' and 'RX' pins in UART communication; compare 'I2C' and 'SPI' in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺപ്റ്റ് കോൺസെപ്റ്റ് കോൺസെപ്റ്റ്. കോൺപ്റ്റ് diagram / circuit / formula / procedure കോൺസെപ്റ്റ്. കോൺസെപ്റ്റ് result കോൺസെപ്റ്റ് application കോൺസെപ്റ്റ് കോൺസെപ്റ്റ് കോൺസെപ്റ്റ്. Technical terms English-ൽ കോൺസെപ്റ്റ് കോൺസെപ്റ്റ്.

Common mistake / precaution: Incorrect communication settings (baud rate, parity) will result in data corruption. All communication links must be verified with authorised diagnostic tools.

2. Design and Simulation–

Embedded system design involves selecting the right components and writing robust software. Tools like Proteus allow for virtual prototyping and debugging before hardware fabrication. In-Circuit Debugging (ICD) helps identify run-time errors on the actual hardware.

Study and application method

Describe the 'Hardware-Software Co-design' process; explain the advantage of using 'Proteus' for embedded system development.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Hardware-Software Co-design' process; explain the advantage of using 'Proteus' for embedded system development.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നോക്കുക. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കുക. ഏതു result ലഭിക്കും. ഏതു application ഉപയോഗിക്കും. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Simulation results may differ from real-world behavior. All designs must be validated on physical hardware using authorised test procedures and safety checks.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: PIC Microcontroller Architecture

Use the concept flow to revise observation, principle, method and verification.

Module 2: Embedded C Programming for PIC

Use the concept flow to revise observation, principle, method and verification.

Module 3: Peripheral Interfacing and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Embedded System Design and Simulation

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare 8051 and PIC18F microcontrollers in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Write a PIC C program to toggle all pins of Port B with a 500ms delay.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the flowchart for a 'Digital Thermometer' system using a PIC microcontroller and LM35 sensor.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the circuit diagram for a 'Password-based Lock' using a 4x4 keypad and 16x2 LCD.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify three household or industrial products that use PIC microcontrollers (e.g., microwave, washing machine).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — PIC Microcontroller Architecture

Explain PIC internal structure and state one correct method, application or precaution.—

PIC (Peripheral Interface Controller) microcontrollers use a Harvard architecture with separate buses for instructions and data. The PIC18F series features an 8-bit CPU, multi-level hardware stack, and integrated peripherals. Pipelining allows fetching and executing instructions simultaneously.

Answer framework: Sketch the block diagram of the PIC18F microcontroller; explain the advantage of 'Harvard Architecture' over Von Neumann.

Important point: PIC microcontrollers are sensitive to ESD. Do not handle the ICs without authorised grounding and anti-static precautions.

Explain Memory and SFRs and state one correct method, application or precaution.—

Program memory stores code in Flash, while Data memory uses SRAM and EEPROM. Special Function Registers (SFRs) like TRIS, PORT, and LAT are used to control I/O and internal modules. The 'Access Bank' simplifies register addressing.

Answer framework: Describe the role of 'TRIS' and 'PORT' registers in I/O control; explain the purpose of the 'EEPROM' in embedded systems.

Important point: Improper memory mapping or stack overflow can cause system crashes. All memory-intensive operations must follow authorised software design guidelines.

Module 2 — Embedded C Programming for PIC

Explain Software development environment and state one correct method, application or precaution.—

MPLAB X IDE is the official development platform for PIC. The XC8 compiler converts C code into machine-level hex files. 'Configuration Bits' are used to set the oscillator type, watchdog timer, and power-up timer before code execution.

Answer framework: Explain the steps to create a new project in 'MPLAB X IDE'; write a C code snippet to configure the internal oscillator.

Important point: Incorrect configuration bits can prevent the microcontroller from starting. Do not program configuration bits without authorised verification of the target hardware setup.

Explain Timers and Interrupts and state one correct method, application or precaution. –

Timers are used for generating precise delays and measuring time intervals. Interrupts allow the CPU to respond to external or internal events immediately. PIC18F supports high and low priority interrupts for efficient task management.

Answer framework: Write a C program to generate a 1-second delay using 'Timer0'; explain the difference between 'Polled I/O' and 'Interrupt-driven I/O'.

Important point: Uncontrolled interrupt nesting can lead to stack overflow. All interrupt service routines (ISRs) must be kept short and follow authorised real-time design rules.

Module 3 — Peripheral Interfacing and Control

Explain User interface and sensors and state one correct method, application or precaution. –

Interfacing LCDs and keypads allows for user interaction. The ADC module converts analog sensor voltages (e.g., from an LM35 temperature sensor) into digital values for processing. Proper voltage scaling and reference selection are essential for accuracy.

Answer framework: Sketch the circuit for interfacing a '16x2 LCD' with PIC18F in 4-bit mode; write a C function to read a value from the 'ADC'.

Important point: Sensor signals are prone to noise. Do not interface sensors without authorised signal conditioning, filtering, and shielded cables.

Explain Actuator control using PWM and state one correct method, application or precaution. –

The Capture/Compare/PWM (CCP) module is used to generate PWM signals. PWM is essential for controlling the speed of DC motors and the brightness of LEDs. H-bridge drivers like L293D are used to interface motors with the microcontroller.

Answer framework: Explain the principle of 'Pulse Width Modulation' (PWM); sketch the interface between a PIC18F and a 'DC Motor' using a driver IC.

Important point: Motors generate inductive back-EMF. Do not connect motors to the microcontroller without authorised flyback diodes and isolation circuits.

Module 4 — Embedded System Design and Simulation

Explain Communication protocols and state one correct method, application or precaution.—

UART is used for simple point-to-point serial communication (e.g., with a PC). I2C and SPI are used for communication between the microcontroller and multiple peripheral ICs like RTCs, sensors, and external memory.

Answer framework: Explain the role of 'TX' and 'RX' pins in UART communication; compare 'I2C' and 'SPI' in a conceptual table.

Important point: Incorrect communication settings (baud rate, parity) will result in data corruption. All communication links must be verified with authorised diagnostic tools.

Explain Design and Simulation and state one correct method, application or precaution.—

Embedded system design involves selecting the right components and writing robust software. Tools like Proteus allow for virtual prototyping and debugging before hardware fabrication. In-Circuit Debugging (ICD) helps identify run-time errors on the actual hardware.

Answer framework: Describe the 'Hardware-Software Co-design' process; explain the advantage of using 'Proteus' for embedded system development.

Important point: Simulation results may differ from real-world behavior. All designs must be validated on physical hardware using authorised test procedures and safety checks.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: PIC Microcontroller Architecture.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Embedded C Programming for PIC.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Peripheral Interfacing and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Embedded System Design and Simulation.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Introduction to PIC microcontrollers; PIC18F architecture: CPU, Harvard architecture, pipelining; Memory organization: Flash (Program), SRAM (Data), EEPROM; Special Function Registers (SFRs); Oscillator and Reset..
2. Develop a complete answer or practical plan for: Embedded C fundamentals; MPLAB X IDE and XC8 compiler; Configuration bits; I/O programming; Timers and Counters; Interrupt handling in PIC18F..
3. Develop a complete answer or practical plan for: Interfacing: LED, 16x2 LCD, 4x4 Keypad; Analog-to-Digital Converter (ADC) module; CCP module: PWM for DC motor control; Sensors: Temperature (LM35), Ultrasonic..
4. Develop a complete answer or practical plan for: Serial communication: UART (RS232); Basics of I2C and SPI; Hardware-software co-design; Proteus simulation; Debugging techniques (ICD); Simple project: Digital thermometer or Speed controller..

LAST-MILE REVISION

Quick Revision

Module 1: PIC Microcontroller Architecture

Introduction to PIC microcontrollers; PIC18F architecture: CPU, Harvard architecture, pipelining; Memory organization: Flash (Program), SRAM (Data), EEPROM; Special Function Registers (SFRs); Oscillator and Reset.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Embedded C Programming for PIC

Embedded C fundamentals; MPLAB X IDE and XC8 compiler; Configuration bits; I/O programming; Timers and Counters; Interrupt handling in PIC18F.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Peripheral Interfacing and Control

Interfacing: LED, 16x2 LCD, 4x4 Keypad; Analog-to-Digital Converter (ADC) module; CCP module: PWM for DC motor control; Sensors: Temperature (LM35), Ultrasonic.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Embedded System Design and Simulation

Serial communication: UART (RS232); Basics of I2C and SPI; Hardware-software co-design; Proteus simulation; Debugging techniques (ICD); Simple project: Digital thermometer or Speed controller.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
PIC internal structure	PIC (Peripheral Interface Controller) microcontrollers use a Harvard architecture with separate buses for instructions and data.
Memory and SFRs	Program memory stores code in Flash, while Data memory uses SRAM and EEPROM.
Software development environment	MPLAB X IDE is the official development platform for PIC.
Timers and Interrupts	Timers are used for generating precise delays and measuring time intervals.
User interface and sensors	Interfacing LCDs and keypads allows for user interaction.
Actuator control using PWM	The Capture/Compare/PWM (CCP) module is used to generate PWM signals.
Communication protocols	UART is used for simple point-to-point serial communication (e.
Design and Simulation	Embedded system design involves selecting the right components and writing robust software.

LEARNING RESOURCES

References

Recommended PIC embedded design references

1. Muhammad Ali Mazidi, PIC Microcontroller and Embedded Systems: Using Assembly and C for PIC18.
2. Tim Wilmshurst, Designing Embedded Systems with PIC Microcontrollers.
3. Dogan Ibrahim, PIC Microcontroller Projects in C.
4. Microchip Technology, PIC18F Family Data Sheet.

Approved public PIC embedded design sources

- <https://nptel.ac.in/courses/108105102>
- https://onlinecourses.nptel.ac.in/noc20_cs66/preview

These sources provide the verified study basis while official Course 6033B detail is unavailable; they do not replace official product designs, authorised safety-critical code, certified equipment specifications or authorised strategic plans.